

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - TECHNICIAN CERTIFICATE

CONTINUOUS ASSESSMENT TEST - I - MAY-2026

CST 05202 - BASICS OF DATA COMMUNICATION NETWORKS

SEMESTER: II

DURATION: 1 Hr 30 Min.

YEAR: II

MAX. MARKS: 50

PART - A (Answer all the Questions) (7 x 2 = 14)

1. What is data communication?
2. Outline the Components of Data Communication.
3. Define the term "Modulation" in data communication.
4. Mention the types of Data Communication.
5. Differentiate the term "Multiplexing and Demultiplexing".
6. How do you know about the word "Framing" as explained in Data Link Layer.
7. Outline the Sub layers which are very special for Data Link Layer.

PART - B (Answer any three Questions) (3 x 4 = 12)

8. The Transmission channel or Medium is a link that carries a message between two or more Devices. Summarize the Categories of Data communication channels available in Data communication.
9. Discuss the Protocols available in the among of the layer known as "Application Layer" found in OSI Model.
10. Briefly explain the concept of Transmitter and Receiver in Data communication.
11. Shortly elaborate the Main Purpose of the Network Layer in the OSI Model.
12. Outline the Techniques used in Error Control for the Data Link Layer.
13. Define the term Routing Algorithms and Mention the Classifications of Routing Algorithms as used in Network layer.

PART - C (Answer any two Questions) (2 x 12 = 24)

14. Reference Model in Computer Networking are Standard frameworks that explain how data is Transmitted and communicated between devices in the Network. Expound the all the layers found in the Open Systems Interconnection.
15. With the vivid examples explain in details about the Error Detection and Error Correction.
16. Explain in Details about the Mobile Network Architecture.
17. Explain in details about the all types of Multiplexing.